

# Nicole Fong

+1 (650) 484-9212 | [nicolefongjw@gmail.com](mailto:nicolefongjw@gmail.com) | [linkedin.com/in/coleng](https://www.linkedin.com/in/coleng) | [github.com/nclfng](https://github.com/nclfng) | [nicolefong.tech](https://nicolefong.tech)

## EDUCATION

### University of California, San Diego

Anticipated Graduation: June 2029

*Bachelor of Science in Artificial Intelligence – Provost Honors, all quarters*

*GPA: 3.78*

- **Relevant Coursework:** Intro to AI, Accelerated Java, Data Structures, Object-Oriented Design, Linear Algebra, Multivariable Calculus, Discrete Math, Math for Algorithms and Systems, Systems Programming & Software Tools

## EXPERIENCE

### CSE PACE Lead Mentor

Sept. 2026 - Present

*UC San Diego, Computer Science & Engineering*

*San Diego, CA*

- Mentor 100+ first-year students in the CSE PACE program by leading small-group activities in a weekly session, building peer community and academic confidence in the major.
- Facilitate hands-on group exercises and discussions each week, connecting first-year students to CSE courses, resources, and campus opportunities.

### Google Student Ambassador

Aug. 2026 - Present

*Google*

*San Diego, CA*

- Represent Google as 1/100 students out of 35,000+ applications chosen to lead campus outreach at 50 U.S. universities.
- Plan and help run on-campus events and outreach initiatives to build awareness of Google's tools, career resources, and student programs.

### AI & ML Intern

June 2026 - Aug. 2026

*ChakraTech*

*San Diego, CA*

- Led development of a property-prediction pipeline for biodegradable polymers, engineering structural features from molecular data to prioritize which candidate materials should move forward to physical testing.
- Identified and corrected a data leakage risk in the modeling approach by grouping structurally related compounds during train/test splitting, preventing inflated accuracy estimates.

### Operations Assistant

Jan. 2026 - Aug. 2026

*Rad Blue Inc*

*San Francisco, CA*

- Improved performance for sales verticals by 18% across 30+ tasks by building ground-truth evaluation baselines and rubric-based assessments for RLHF fine-tuning.

### AI Events Board Director & AI Projects Mentor

Oct. 2025 - Present

*Association for Computing Machinery (ACM)*

*San Diego, CA*

- Direct AI event programming for UCSD's largest student organization, planning 5+ events this fall including a 4-part AI School Series workshop track and the AI Fall Kickoff.
- Mentored 15+ students across three cohorts on applied ML fundamentals using PyTorch, scikit-learn, and Hugging Face.

### Co-President

Oct. 2025 - Present

*Triton Engineering Student Council (TESC)*

*San Diego, CA*

- **Co-President (May 2026 - Present):** Leading the umbrella organization for 50+ UCSD engineering clubs, setting fall priorities for club funding and corporate sponsorships, and hosting events for 500+ students.
- **Tech Committee (Oct. 2025 - May 2026):** Reduced manual data entry by 90% by building a Python-based scraper to automate engineering club data collection, and managed technical event logistics and digital check-ins for 100+ attendees.

## PROJECTS

### DiamondHacks: Straightline | *Gaussian Splatting, BrowserUse API, Supabase, Python*

Apr. 2026

- Won 1st place (Best Use of Browser Use) among 400+ participants at DiamondHacks 2026 by building autonomous browser agents to audit ADA compliance across live web interfaces, structuring audit data into a Supabase cache layer that fed the team's 3D Gaussian Splatting accessibility visualization.

### Triton Robotics Autonomy Team Capstone | *ROS2, OpenCV, C++, Python*

Oct. 2025 - Mar. 2026

- Achieved 6-DoF pose estimation of armor targets by replacing bounding-box detection with feature-based keypoint matching, applying solvePnP over 2D-3D correspondences with measured object geometry.
- Prototyped the OpenCV contour-detection node in Python and integrated it into the team's ROS2 perception stack.

### ACM AI Projects: Fruit Recognition Model | *PyTorch, YOLO, Python*

Oct. 2025 - Jan. 2026

- Trained a 24-layer GoogLeNet-inspired CNN in PyTorch on COCO 2017 by implementing the Inception modules and auxiliary classifiers from the original paper.
- Built IoU computation and a multi-part YOLO-style loss (localization, objectness, classification) from scratch without detection libraries.

## TECHNICAL SKILLS

**Languages & Tools:** Python, C++, Java, TypeScript, Swift/SwiftUI, HTML/CSS, Linux, Git

**Computer Vision & Robotics:** OpenCV, PyTorch, CNNs, YOLO, solvePnP, ROS2

**Machine Learning & AI:** Model Evaluation, RLHF, Hugging Face, scikit-learn, pandas, NumPy, RDKit, PSMILES

**Web Development:** React, Next.js, Supabase, Tailwind CSS, BeautifulSoup, BrowserUse API